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Members' Name	Total Group Mark	Individual Mark	Total Mark
1) SHANJAAY			
2) DHIVIYAN			
3) WONG JIA JUN			
4) LOH MIN YU			

Date of Submission : 17/3/2025

Date received :

Table of Contents

Part 1: Understanding the OSI Model and Network Layer 3
a) Layer Identification in a Real Network Setup: 3
b) Real-Life Application Scenario 5
Part 2: Types of Networks and Quality of Service (QoS) 7
c) Different Type of Networks: 7
d) Explain QoS and its important in networking: 8
e) Quality of Service (QoS) Strategies: 9
Part 3: Cisco Packet Tracer Simulation 10
f) Examine HTTP Web Traffic 10
g) Display Elements of the TCP/IP Protocol Suite 14
References 18

Part 1: Understanding the OSI Model and Network Layer

a) Layer Identification in a Real Network Setup:

The Open Systems Interconnection (OSI) model consist seven layers including Physical, Data Link, Network, Transport, Session, Presentation, and Application layer. Different devices and protocols interact with each layer in a home network to ensure smooth and efficient data transfer. The below explanation describes a typical home network OSI model mapping.

1. **Physical Layer**

Modems, routers, switches, and network adapters are example of physical devices used for data transmission in the physical layer (Chrystal, 2024). Additionally, cables like fiber optic cables and Ethernet cables (Cat6, Cat7) are needed for data transfer and to connect network devices. Most commonly, these two types of cable used in home network to connect and configure network devices. For instance, in a home network, fiber optic cable is connected to a modem, which is connected to a router using an Ethernet cable. After then, the router transmits data to devices like laptops and smartphones via wired Ethernet connections or wireless Wi-Fi connections (Parrish, 2024).

2. **Data Link Layer**

Devices on the same network can communicate reliably due to the Data Link layer. It efficiently manages data transport and identifies devices using MAC (Media Access Control) addresses. The protocol for Ethernet is IEEE 802.3, and for Wi-Fi is IEEE 802.11 which handles data communication within the network (Rosenbaum, 2022). For example, when a laptop connects to the router via Wi-Fi, the router checks the MAC address of the device before allow network access. Then the user in the home can use the connected laptop to send an email or browse a website.

3. **Network Layer**

The Network layer is responsible for IP addressing (IPv4 or IPv6), packet forwarding, and routing between networks. The router serves as a gateway in a home network, connecting

local devices to the internet. With the use of DHCP (Dynamic Host Configuration Protocol), the router provides home devices with private IP addresses such as 192.168.1.x (GeeksforGeeks, 2024). For example, default IP address for home router can be 192.168.1.1. Then, it will dynamically assign 192.168.1.2, 192.168.1.3, etc to other locally connected devices like laptops and smartphones (Fortinet, n.d.). ISP (Internet service Provider) assigns a public IP address to the router, so that local devices can communicate with external networks like Internet through router (Fortinet, n.d.).

4. **Transport Layer**

   The Transport layer uses TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) to control data delivery and enable end-to-end communication. The TCP is a reliable protocol used for web browsing (HTTP/HTTPS), emails, and file downloads. It also ensures data is received as whole and in correct order using error-checking and retransmission. In terms of UDP, it prioritize speed and efficiency, but less reliable because of its connectionless protocol that does not guarantee delivery, sequence, or error-free transmission of data. It is mostly used for video streaming and gaming. While TCP request for retransmission if packets are lost, but UDP does not resend lost packets (GeeksforGeeks, 2024). For example, watching a movie on Amazon Prime in a smart TV uses UDP for smooth playback, while sending an email uses TCP to ensure complete data transmission.

5. **Session Layer**

   Active communication sessions between devices are controlled and maintained by the session layer. It guarantees that while data is being transferred, sessions stay open and can be re-established as necessary. In a home network, a Microsoft Teams meeting keeps the connection between the user's laptop and the Team's server open. The session layer automatically reconnects without having to restart the meeting if the connection fails.

6. **Presentation Layer**

   For a safe and effective communication, the Presentation layer manages data formatting, encryption, and compression. To format the data, the layer will utilize data encoding methods like ASCII and Unicode so that it can be understood by computers. SSL (Secure Sockets Layer) / TLS (Transport Layer Security) ensures encrypted connections for safer

data transmission. Moreover, HTTPS in a URL shows a secured connection, thus it is secure for accessing the websites. In addition, presentation layer compresses data for faster transmission. For instance, when using HTTPS to compose email in Gmail (web version), SSL/TLS encrypts the data before sending it over the network.

7. **Application Layer**

The application layer allow users to directly access to data through software applications. In application layer, there will be many protocols the application will used depends on the task performed, data being transferred, and applications functionality. For example, for web browsing HTTP or HTTPS protocols will be used to access websites. FTP (File Transfer Protocol) is used for uploading or downloading files to or from local servers or external storage in a network. Apart from that, DNS (Domain Name System) is used to convert domain names like google.com or youtube.com into machine understandable IP addresses.

b) Real-Life Application Scenario

Sending an email:

1. **Application Layer (Layer 7)**

Application Layer is the layer that enable users to compose email and send it to another user using the user interface in an email client such as Gmail, Microsoft Outlook, and Yahoo!. For sending an email the application layer will determine the required protocol. SMTP (Simple Mail Transfer Protocol), POP (Post Office Protocol), and IMAP (Internet Message Access Protocol) are the three email protocols used to send and receive email (GeeksforGeeks, 2022). SMTP's MIME (Multipurpose Internet Mail Extension Protocol) extension protocol enable the delivery of non-ASCII data including photos, videos, and audios (GeeksforGeeks, 2022). Once the necessary protocols are determined and applied, the email data is passed to the Presentation layer for further processing.

2. **Presentation Layer (Layer 6)**

The Presentation layer ensures that the data is formatted correctly so that it can be understood by the receiving system. Thus, it is responsible for translation, encryption, and data compression. Therefore, presentation layer will translate the email content into a standard encoding such as ASCII (American Standard Code for Information Interchange) or Unicode. The email content will be encrypted for safer transmission using TLS (Transport Layer Security) or SSL (Secure Sockets Layer) email encryption if encryption is enabled (Google, n.d.). To boost speed and efficiency, the layer compresses the email to reduce the size of the data before forwarding it to the next layer.

3. **Session Layer (Layer 5)**

Session layer will start a new communication session between the sender's and receiver's email servers. It manages and maintains the session, ensuring that the connection remains open until the entire email is sent and received by the receiver.
After ensuring a stable connection, the data is sent to the Transport layer.

4. **Transport Layer (Layer 4)**

The email data will be segmented into smaller pieces for efficient transmission over the network. Transmission Control Protocol (TCP) is used and TCP header will be added in each segment, which includes source and destination ports, acknowledgement number and error checking details (GeeksforGeeks, 2024). Moreover, each segment will be assigned a sequence number to be reassembled correctly and in proper order at the destination. If there is any lost packets, it will request for retransmission.

5. **Network Layer (Layer 3)**

During the Network layer, each segment from Transport layer will be encapsulated into packets. The Internet Protocol (IP) is used to add sender's and receiver's IP addresses, allowing the email data to be routed across multiple networks and reach the correct destination. Then, the packets are forwarded to the Data Link layer.

6. **Data Link Layer (Layer 2)**

The packets will be encapsulated into frames by adding sender's and receiver's MAC (Media Access Control) addresses upon received from the Network layer. This will ensure proper delivery across the local network. The Data link layer decides whether the data is sent over a wired network like Ethernet or wireless network (Wi-Fi). But commonly email is sent through Wi-Fi. The frames are passed to the Physical layer for transmission.

7. **Physical layer (Layer 1)**

The frames are converted into binary bits (0s and 1s). These bits are then transmitted as electrical signals over Ethernet cables, fiber-optic cables, or radio waves in wireless communication like Wi-Fi (GeeksforGeeks, 2025). The data travels across the internet to reach the recipient's email server.

Then, the process of receiving email will happen at the recipient's end. Starting from Physical layer to the Application layer in the OSI model.

Part 2: Types of Networks and Quality of Service (QoS)

c) Different Type of Networks:

Local Area Network

A LAN is a simple and small network where two or more devices are connected to share data and resources within a limited geographical area. For example, an office network where computers and devices in an office are connected to the same router or switch to share files and printers. Another example is a home network where a Wi-Fi router connects laptops, smartphones, and smart devices within a house (Peng, 2018).

Wide Area Network

A WAN is a network that connects different LANs from different geographical areas into one single network. For example, banking networks where a bank's branches in different cities or countries use a WAN to access centralized databases. Corporate networks also use

WANs to connect offices across different locations using leased lines or VPNs (Peng, 2018).

Metropolitan Area Network

A MAN is a network that connects computers within a metropolitan area, which could be a single large city, multiple cities, or any given large area with multiple buildings. For example, cable TV networks use MANs to distribute TV channels and internet services across a city or large metropolitan area. They use fiber-optic or coaxial cables to connect different neighborhoods to a central broadcasting station (Peng, 2018).

d) Explain QoS and its important in networking:

Quality of Service (QoS) refers to a set of network management techniques which ensure optimal performance for different types of data traffic. In IoT environments, QoS helps to enable efficient communication between devices and prevents network congestion.

Importance of QoS in Networking

Traffic Prioritization

QoS allows for classifying network traffic, ensuring that time-sensitive applications like video conferencing and VoIP receive higher priority over less critical tasks like file downloads and web browsing.

Bandwidth Management

As network resources are limited, QoS optimizes bandwidth allocation to prevent waste and improve overall network performance.

Latency and Jitter Reduction

Jitter refers to variations in packet arrival times, which can degrade the quality of voice and video communication. QoS minimizes jitter by adjusting packet scheduling.

Packet Loss Reduction

Packet loss can negatively impact communication quality. QoS ensures data integrity through error correction and retransmission mechanisms. (Singh,2018)

e) Quality of Service (QoS) Strategies:

Traffic Classification and Marking

QoS classifies and marks network traffic using techniques such as Differentiated Services Code Point (DSCP) and Class of Service (CoS). This allows routers and switches to identify high-priority packets and process them accordingly.

Queuing Mechanisms

Various queuing methods help manage traffic efficiently. First In, First Out (FIFO) processes packets in order of arrival, while Priority Queuing (PQ) gives preference to critical traffic such as voice and video. Weighted Fair Queuing (WFQ) allocates bandwidth fairly among different types of network traffic.

Congestion Management

In order to prevent excessive delays, QoS applies congestion control methods such as Random Early Detection (RED) which drops low-priority packets when the network approaches congestion levels. Traffic shaping techniques regulate data flow to prevent sudden traffic spikes.

Resource Reservation

Certain applications require dedicated bandwidth to function correctly. Integrated Services (IntServ) uses the Resource Reservation Protocol (RSVP) to allocate bandwidth while Differentiated Services (DiffServ) categorizes traffic into service classes to ensure the quality transmission.

Example:

Netflix is one of the world's largest video streaming applications which is needed to deliver high-quality content to millions of users globally. In order to ensure smooth playback

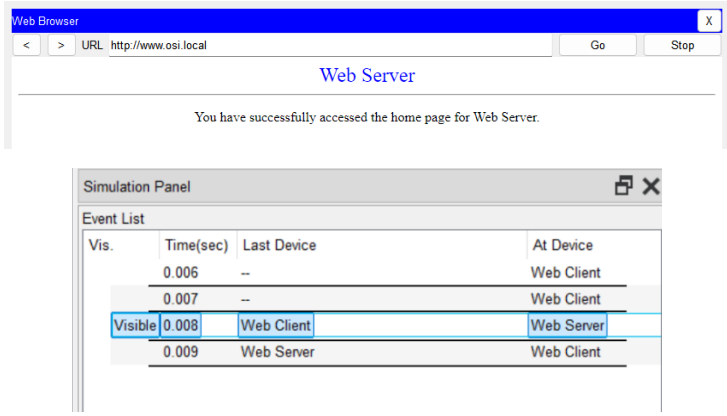
without the buffering, Netflix relies on Quality of Service (QoS) mechanism that optimize network performance by prioritizing streaming traffic and managing bandwidth efficiently. Netflix will use the Adaptive Bitrate Steaming (ABS) which will dynamically adjusts video quality based on available bandwidth to ensure the users can receive the highest possible resolution. (Singh, 2018)

Part 3: Cisco Packet Tracer Simulation

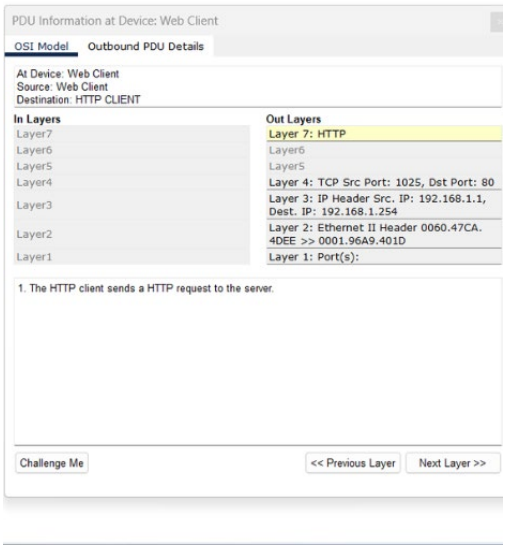
Investigate the TCP/IP and OSI Models in Action

f) Examine HTTP Web Traffic

- **Look at the Web Client web browser page. Did anything change?**
  - There are 4 records in the event list and the web browser shows "you have successfully accessed the home page for Web Server."

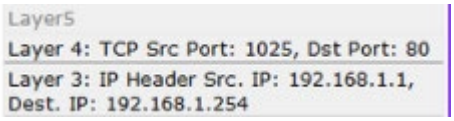


- **What information is listed in the numbered steps directly below the In Layers and Out Layers boxes for Layer 7?**



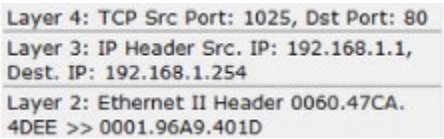
- 1. The HTTP client sends a HTTP request to the server.

- **What is the Dest Port value for Layer 4 under the Out Layers column?**



- The Dest Port value is 80.

- **What is the Dest. IP value for Layer 3 under the Out Layers column?**



- The Dest IP value for it is 192.168.1.254.

- **What information is displayed at Layer 2 under the Out Layers column?**

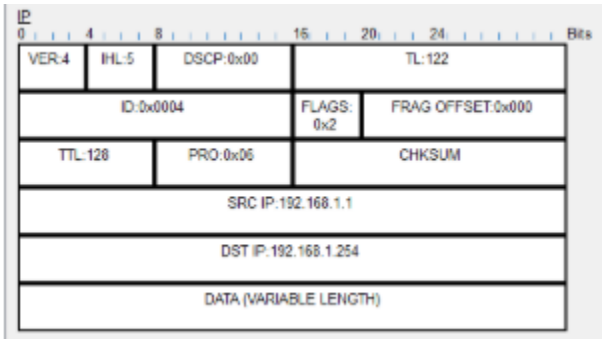
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer 1: Port(s):

1. The next-hop IP address is a unicast. The ARP process looks it up in the ARP table.

2. The next-hop IP address is in the ARP table. The ARP process sets the frame's destination MAC address to the one found in the table.

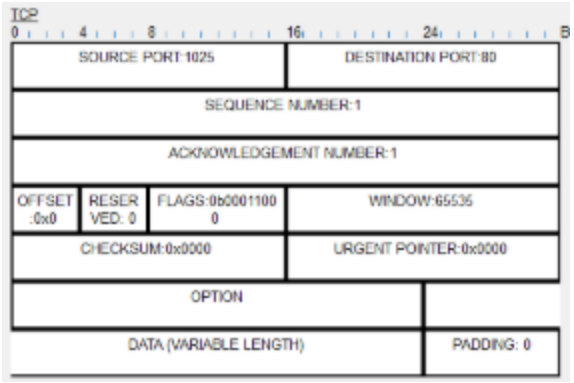
3. The device encapsulates the PDU into an Ethernet frame.

- **What is the common information listed under the IP section of PDU Details as compared to the information listed under the OSI Model tab? With which layer is it associated?**



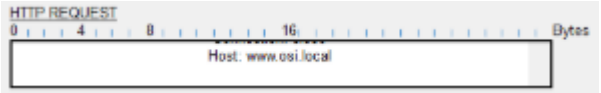
- The common information listed under IP section of PDU details includes source IP address (192.168.1.1) and destination IP address (192.168.1.254), associated with layer 3 (network layer) of the OSI model.

- **What is the common information listed under the TCP section of PDU Details, as compared to the information listed under the OSI Model tab, and with which layer is it associated?**



- The common information listed under TCP section of PDU details includes source port (1025) and destination port (80), that associated with layer 4 (transport layer) of the OSI model.

- **What is the Host listed under the HTTP section of the PDU Details? What layer would this information be associated with under the OSI Model tab?**



- The Host listed under the HTTP section of the PDU Details is www.osi.local. The information associated with under the layer 7 (application layer) of the OSI model.

- **Comparing the information displayed in the In Layers column with that of the Out Layers column, what are the major differences?**
  - Comparing the In Layers column with the Out Layers column, the major differences are source and destination address and flow direction.

- **How many tabs are displayed with this event? Explain.**

- The PDU information window displays two tabs which are OSI Model and Inbound PDU details. We can analyze the details of the packet as it moves through the network. the OSI Model tab displays the encapsulation layers of the packet showing protocol information for each layer. The Inbound PDU Details tab provides more information about the packet transmission process and the details of the data transmission.

g) Display Elements of the TCP/IP Protocol Suite

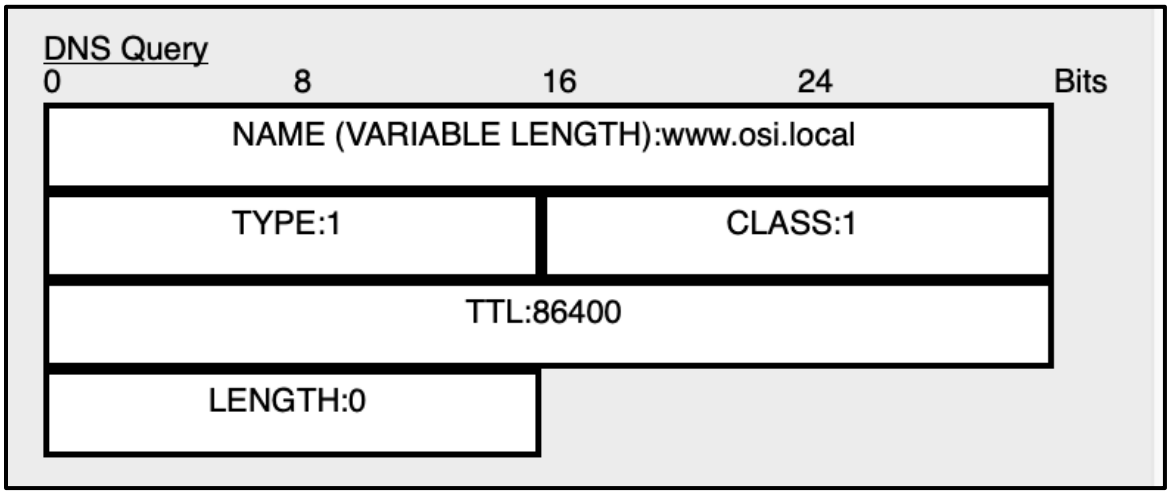
- **What additional Event Types are displayed?**

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	Web Client	DNS
	0.000	--	Web Client	ARP
	0.001	Web Client	Web Server	ARP
	0.002	Web Server	Web Client	ARP
	0.002	--	Web Client	DNS
	0.003	Web Client	Web Server	DNS
	0.004	Web Server	Web Client	DNS
	0.004	--	Web Client	TCP
	0.005	Web Client	Web Server	TCP
	0.006	Web Server	Web Client	TCP
	0.006	--	Web Client	HTTP
	0.007	Web Client	Web Server	TCP
	0.007	--	Web Client	HTTP
	0.008	Web Client	Web Server	HTTP
	0.009	Web Server	Web Client	HTTP
	0.009	--	Web Client	TCP
	0.010	Web Client	Web Server	TCP
	0.011	Web Server	Web Client	TCP
	0.012	Web Client	Web Server	TCP

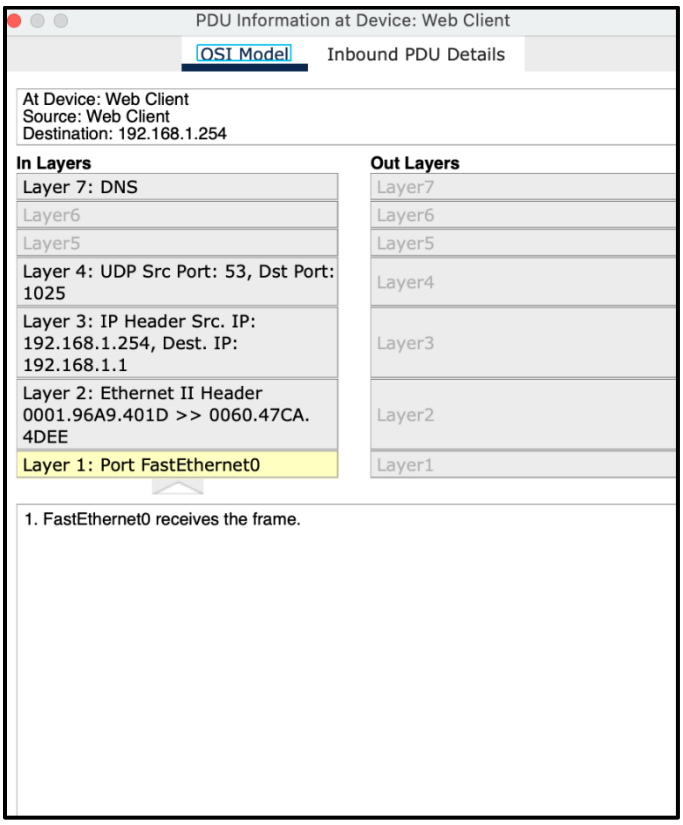
- DNS, ARP AND TCP

- **What information is listed in the NAME field: in the DNS QUERY section?**



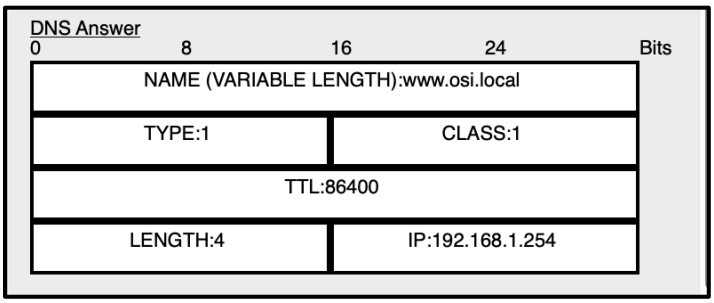
- www.osi.local

- **At which device was the PDU captured?**



- Web Client

- **What is the value listed next to ADDRESS: in the DNS ANSWER section of the Inbound PDU Details?**



- 192.168.1.254.

- **In the numbered list directly below the In Layers and Out Layers, what is the information displayed under items 4 and 5?**

1. The device receives a TCP ACK segment on the connection to 192.168.1.1 on port 1025.
2. Received segment information: the sequence number 1, the ACK number 1, and the data length 20.
3. The TCP segment has the expected peer sequence number.
4. The TCP connection is successful.
5. The device sets the connection state to ESTABLISHED.

- The TCP connection is successful
- The device sets the connection state to ESTABLISHED

- **What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?**

1. The device receives a TCP ACK segment on the connection to 192.168.1.1 on port 1025.
2. Received segment information: the sequence number 104, the ACK number 273, and the data length 20.
3. The TCP segment has the expected peer sequence number.
4. The device sets the connection state to CLOSED.

- To terminate the TCP connection.

Challenge Questions

This simulation provided an example of a web session between a client and a server on a local area network (LAN). The client makes requests to specific services running on the server. The server must be set up to listen on specific ports for a client request. (Hint: Look at Layer 4 in the OSI Model tab for port information.)

• Based on the information that was inspected during the Packet Tracer capture, what port number is the Web Server listening on for the web request?

- The Web Server is listening on port 80 for the web request.
- This can be seen in Layer 4 of TCP where the destination port is 80, which is the standard port for HTTP web traffic.

• What port is the Web Server listening on for a DNS request?

- The Web Server is listening on port 53 for DNS requests.
- In the earlier PDU captures, the UDP segment had a source port of 53, indicating that the server is handling DNS queries on this port.
- Port 53 is the standard port for DNS services.

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